

Chevron Scholarship

PROGRAM DEADLINE : June 04 2023 at 11:59 PM PDT (Midnight)

Applicant Information

Name: Jacob Vaught
Email: jvaught@email.sc.edu
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Applicant Information

Personal Information

First Name	Jacob
Last Name	Vaught
Email Address	jvaught@sc.edu
Phone Number	803-645-9962
Date of Birth	07/13/2003
Street Address	211 Main St
Street Address 2	Apt 111
City	Columbia
State/Province	SC
Zip Code	29201
Country	United States
Ethnicity. Please select all that apply.	Two or more races

Which identities do you hold? Select all that apply.	None of the above
Are you employed full-time or part-time?	Part-time
Are you a U.S. citizen or permanent resident? *Citizenship is <i>not</i> a requirement of SHPE-funded scholarships; however, some corporate-funded scholarships may require citizenship.	Yes
Gender	Male

SHPE Information

Are you an active SHPE member?	Yes
Current Membership Type	SHPE Student Member
Membership ID #	1889156
In which region is your chapter?	7
Do you plan on attending the 2023 National Convention in Salt Lake City, UT?	Yes

Financial Information

Are you eligible to receive Pell Grant in Fall 2023?	Yes
Did you participate in the free and reduced lunch program during your childhood?	I don't know
What is your SAI (Student Aid Index) from your 2022 or 2023 FAFSA?	
<i>If you have not yet completed the 2023 FAFSA, please provide your SAI from your 2022 FAFSA.</i>	00000
Leave the response blank if you have not completed a 2022 nor a 2023 FAFSA.	
Are you currently employed full-time in a STEM related field?	No
Employer	

Supervisor Full Name

Supervisor Phone Number

Academic Information

High School Academic Information

High School Attended Homeschooled

City of High School Aiken

State of High School SC

High School Graduation Date 05/2021

College/University Academic Information

Degree Pursing during the 23-24 Academic Year Bachelor

Major Electrical Engineering

Are you majoring in more than one subject? Yes

What is your other major? Computer Science

Academic Status for the 23-24 Academic Year Junior

Which college/university will you be attending during the 23-24 academic school year? University of South Carolina

State in which your college/university is located SC

GPA & Transcripts

Current GPA 3.56

Please upload your transcript - unofficial is acceptable. Vaught_Jacob_Y10786492_12_11_2022.pdf

Scholarship Questions

Response Questions

Summarize your life experiences and any challenges that have impacted your path to higher education.

As a homeschooled child, I struggled with a severe speech impediment and lacked writing skills when I entered university, due to my non-English speaking mother avoiding English/grammar while schooling us. Despite these challenges, I remained determined to overcome them and worked hard to improve my writing skills and academic performance.

However, my path to higher education has been further impacted by personal circumstances. After reporting the abuse of my younger sisters by my father, my parents disowned me, leaving me without any financial support or guidance. To make ends meet, I work part-time conducting research and also TA classes to pay for my housing and balance it with my academic responsibilities.

Despite these challenges, I remain committed to my academic and professional goals, as well as my passion for preventing abusive situations in the future. As a student majoring in Computer Science, Electrical Engineering, and Mechanical Engineering, I am passionate about utilizing my skills to improve the world and prevent abusive situations in the future. My involvement in various engineering clubs, serving as a senator, and teaching assistant demonstrates my commitment to advancing my knowledge and making a positive impact on campus. I have also volunteered with organizations that address and prevent domestic violence and child abuse.

Any form of financial support would be crucial in enabling me to continue pursuing my academic and professional aspirations. I would be able to devote more time and energy to my studies and to making a positive impact in the world.

Discuss your motivation for pursuing a STEM degree.

From a young age, I have been deeply passionate about the potential of science and technology to transform society and make a positive impact on the world. Pursuing a degree in Electrical Engineering, Computer Science, and Mechanical Engineering (EECS/ME) has allowed me to channel this passion in a meaningful way and work towards a future where technology is used to benefit everyone.

Through my studies, I have developed a deep understanding of the principles and techniques that underlie modern technology and engineering. From programming languages to circuit design and analysis to mechanics and materials science, I have gained a mastery of the technical skills that are essential to success in these fields.

But it is not just technical skills that I have gained through my studies in EECS/ME. I have also developed critical thinking, problem-solving, and teamwork abilities that are essential to success in any field. I have learned to approach complex problems with a methodical and analytical mindset, and I have developed the ability to work collaboratively with others to achieve shared goals.

What drives me most, however, is the desire to make a real difference in the world. I have always been deeply empathetic and attuned to the challenges that people face on a daily basis, and I believe that technology has the power to address many of these challenges in innovative and effective ways. Pursuing a degree in EECS/ME allows me to work towards this goal and use my skills and knowledge to improve people's lives.

Describe your volunteer or community experience with SHPE or other organizations.

As a member of the Society of Hispanic Professional Engineers (SHPE), I have been involved in coordinating the 2nd Annual CEC Diversity Gala, a professional networking conference that celebrates diverse student organizations within the College of Engineering and Computing. Through this experience, I had the opportunity to work with AOE, SWE, WiC, NSBE, Pi Tau Sigma, and ASME (serving as Vice President) to help acquire funding, outreach to professionals, and facilitate interactions between students and professionals. The event was a great opportunity to showcase my leadership skills, learn from inspiring speakers, and connect with potential mentors and employers.

The experience reinforced my commitment to promoting diversity and inclusion in STEM fields and society as a whole. It was inspiring to see how people from all backgrounds could come together to direct a diversity-focused event, and the event furthered our personal and professional growth.

I believe that promoting diversity and inclusion is crucial in all aspects of my life and career, and I am committed to continuing my involvement with SHPE and other organizations that share this goal. Through my volunteer and community service experiences, I have learned that diversity and inclusion can make a positive impact on STEM fields and society, and I am excited to continue making a difference.

Overall, my involvement with SHPE and the 2nd Annual CEC Diversity Gala has been an incredibly rewarding experience that has allowed me to make a positive impact in my community and foster meaningful connections with professionals and peers alike.

How will a ScholarSHPE impact your life and education?

As an active member of SHPE, I am beyond thrilled at the prospect of receiving the ScholarSHPE scholarship. This scholarship would not only make a significant impact on my life and education, but it would also mean the world to me.

Working part-time to pay for housing and tuition has been an incredibly challenging balancing act, and the financial pressures have taken a toll on me both mentally and emotionally. The ScholarSHPE would provide me with the financial support to pursue my academic pursuits without the added stress of worrying about finances. It would also provide me with the freedom to focus on my passions.

The financial support from the ScholarSHPE scholarship would also give me the opportunity to take on more leadership roles within SHPE and other diversity-focused organizations. As someone who is deeply committed to promoting diversity and inclusion in STEM fields, this scholarship would allow me to develop my leadership and networking skills, and make a difference in my community.

The validation and recognition that comes with the ScholarSHPE scholarship would mean everything to me. It would motivate me to continue pursuing my academic and professional goals with even more dedication and determination. The recognition of my hard work and commitment to promoting diversity and inclusion in STEM fields would give me the confidence to keep making a positive impact. It would alleviate the financial pressures I face, provide opportunities for growth and development, and validate my commitment to promoting diversity and inclusion in STEM fields.

Student Profile

Upload your resume.	Jacob Vaught Resume-04-02-2023.pdf
Upload a professional head-shot photo.	IMG_4637.jpg
Would you like to showcase a project or accomplishment?	Yes
Upload any projects that you would like to share.	Portfolio of Projects.pdf

Chevron Questions

Essay Questions

Tell us a little about yourself (include work history, hobbies, and other interests).

Hello! My name is Jacob Vaught, and I am a passionate student and engineer from the University of South Carolina. My academic interests are quite broad and include electrical engineering, mechanical engineering, and computer science, but my curiosity extends far beyond the classroom. I am very fascinated by the intersections of different fields and thrive on exploring unfamiliar territories.

I've been lucky enough to work under some incredible professors, each project with them being an exciting expedition into new knowledge territories. I've had the opportunity to take part in varied research groups, including the Advanced Real Time Systems Laboratory, the Additive Manufacturing Laboratory, and the Public Health Research Center. My time in these groups has allowed me to delve into areas such as finite element analysis modeling, additive manufacturing, structural health monitoring, real time systems, nuclear magnetic resonance, and my most recent addition, Heart rate monitoring and energy expenditure calculation. I was also awarded the Magellan Scholar award in Spring 2023 for my work on the development of a finite element model to predict residual stress and distortion in laser powder bed fusion additive manufacturing.

For those who've had the pleasure of knowing me deeply, they can attest that one of my greatest passions is being a mentor and imparting knowledge to others. I have done so as a Teaching Assistant for Mechatronics and Thermodynamics at the University of South Carolina, and I've relished the thrill of watching students have their own "lightbulb" moments.

In a similar vein, I have a knack for leadership, currently serving as a Student Senator, Vice President of the American Society of Mechanical Engineers, Secretary of Pi Tau Sigma, and having served as co-founder and senior advisor of the USC Rocketry Club, President of the Institute of Electrical and Electronics Engineers, and President of the Society of Automotive Engineers.

Other than my academic research experience, I had the chance to intern at the renowned Westinghouse Electric Company where I honed my skills in computer-aided design, data analysis, and report writing, all while making some great friends along the way.

Outside academics and work, I believe in active participation in campus life. I am likely to be spotted on campus in a button-up shirt, khakis, and dress shoes, often paired with an interesting tie to express my unique style. I'm also a Star Wars fan and sometimes wear a Star Wars tie if I'm feeling feisty.

Other than my interesting sartorial choices, I'm often spotted in the library reading the classics in literature, and I am currently reading the book "Brave New World" by Aldous Huxley. You may also catch me browsing the special collections section of the library on some days, as I am also particularly interested in History, and I have spent many a night researching a person or building I am interested in.

One of the more interesting things that I have spent a lot of time outside of school or work on is my fascination with the human psyche and psychology. I don't do anything with my knowledge here, but I do use things I learn from this to improve my teaching skills.

What are your educational/academic goals? What influenced your choice of major?

Looking at my academic path ahead, I see a clear roadmap: to obtain three bachelor's degrees within five years at USC. These ambitions aren't simply about the accumulation of degrees; they embody a thirst for knowledge and mastery within their respective domains. Within the halls of academia, I aim to contribute original research, with multiple papers published under my name prior to concluding my undergraduate. My long-term vision, however, extends beyond this. I aim to secure a PhD and carve out a niche for myself in the fast-paced and continually evolving fields of robotics and mechatronics.

Reflecting on my academic history, I find a rich tapestry of experiences that has led me to where I am today. It started in a technical college with a Mechatronics program. As I approached the completion of my Associate's degree, I was captivated by the theoretical underpinnings of the machines we built. Inspired, I sought to delve deeper by branching into Mechanical Engineering and Electrical Engineering, the subjects I believed represented Mechatronics.

Upon transitioning to full-time university studies, I had an epiphany: Computer Science was a missing piece of the puzzle. Though I was close to graduating, I knew I needed to include Computer Science in my studies to ensure a meaningful career in Mechatronics.

During my studies, I started research work with a Mechanical Engineering professor. This experience kindled an interest in a graduate degree, which was reinforced after a conversation with an Electrical Engineering professor about the benefits of such a pursuit.

What are your future career plans (include short-term and long-term goals)? Please discuss how the Chevron Scholarship could help you meet these goals.

As I chart my future career path, I envision myself becoming a significant contributor to the global pool of knowledge. My long-term aspirations revolve around flourishing as a proficient researcher, unearthing novel and practical insights that will enrich the world. Beyond that, I hope to be a mentor or educator, empowering and inspiring others to tread a similar path and make their own valuable contributions to society.

In the short term, I am contemplating three distinct career trajectories. I'm considering the prospect of immersing myself in corporate research, developing groundbreaking innovations within a company's ecosystem. Alternatively, I'm exploring the possibility of stepping into a managerial role where I can exercise my leadership skills while further developing my technical acumen. Lastly, I'm weighing the option of an academic career in teaching, nurturing the next generation of innovators.

The Chevron Scholarship stands as a potent catalyst that could help propel me towards these objectives. Firstly, this scholarship would provide the critical financial support I need, enabling me to channel my focus on completing my degree without the burden of financial anxieties.

Secondly, the scholarship would empower me to engage more freely in uncompensated research opportunities allowing me to further refine my capabilities as a researcher, thereby expanding my skills and sharpening my expertise.

Lastly, the scholarship could open doors within Chevron itself. As a prestigious multinational company, Chevron provides an environment where I can leverage my skills and learn from experts in the field and potentially gain a future role there.

Tell us about your previous leadership experiences, whether in high school, college, or during your military service. How have you carried these experiences forward in the classroom, other university activities or internships?

My leadership journey began at home. As the eldest child in a homeschooled family, I led my siblings in tasks, aided with homework, and assisted my parents in lesson planning. At USC, I found confidence in my abilities, leading group projects, tutoring peers, and actively participating in ASME's TigerBurn project where I provided resources and design insights. My contributions led to my election as secretary, where I modernized the ASME lounge and achieved a record recruitment of 50 members in one semester.

This was a springboard to leadership roles in clubs like IEEE, SAE, SAMPE, Rocketry Club, Pi Tau Sigma, and Student Government. As a co-founder of Rocketry Club, I arranged the schedule, spearheaded marketing efforts, and liaised with the school for logistics. My passion for vehicles led to the revival of the SAE automotive racing club, increasing membership to 20 in a single semester. Independently, I ran for and secured a senate position in student government, serving on both the academics and finance committees.

In the workplace, I assumed the lab manager role at ARTS-Lab, implementing measures for accountability and optimizing lab layout for increased student usability. Further, I rearranged office spaces to better accommodate researchers' needs. As an intern at Westinghouse, I initiated and implemented a Python program automating the grid expansion calculation for the proposed switch to 24 from 18-month cycles. My leadership philosophy revolves around innovation, inclusivity, and making a tangible impact in every role I take.

In your opinion, why is teamwork important? What are the key elements you have used to build strong relationships?

I've always believed that teamwork is the bedrock of success, proving the axiom 'the whole is greater than the sum of its parts' to be true. It facilitates diversity of thought, shared responsibility, and enhanced productivity.

When I served as the ASME secretary, I fostered shared responsibility among team members, delegating tasks and promoting collective ownership. This approach led us to welcome 50 new members in a single semester, reaffirming my belief in the power of shared responsibility within a team.

Building strong relationships within teams, in my experience, revolves around three fundamental elements: clear communication, respect, and trust. As the ARTS-Lab manager, I observed firsthand the importance of transparent communication. By clearly defining roles and expectations, providing regular feedback, I fostered an efficient and effective work environment.

While reviving the automotive racing club on campus, I placed respect at the heart of our interactions. We established a culture where every voice was heard and valued, playing a significant role in reenergizing the club and swelling its ranks within a semester.

Trust, however, is the cornerstone of any relationship. During my internship at Westinghouse, I made trust-building with my team a priority. I upheld my commitments, proved my reliability, and stood up for my team members, consequently fostering a trustful team environment. This trust was instrumental in the successful implementation of a program I developed for automating grid expansion calculations. Trust, I've learnt, is the most vital catalyst in the success of any team endeavor.

Tell us about your proudest or most important accomplishment.

As Lab Manager at ARTS-Lab, one accomplishment that resonates with me is transforming our work environment to heighten efficiency and productivity. This was a collaborative effort with my team and the outgoing lab manager as we together restructured the lab's physical space, creating a conducive environment that better suited our needs.

Yet, the challenge was not only physical reorganization but also in introducing a novel machine learning model to track working hours to distribute funds more equitably. The proposal was initially met with some hesitation and resistance. However, I firmly believed in the transformative power of this innovation and the potential it held for our team. I also recognized that change, especially one involving advanced technology, can be daunting.

With my conviction in the idea and the strength of teamwork, I initiated open dialogues to address concerns, answer questions, and gather feedback. I wanted every team member to feel heard, respected, and involved in this change. As a result, we were able to collectively embrace the new system, turning initial pushback into a shared victory.

The outcome was a substantial increase in lab throughput and a higher level of satisfaction among team members. This accomplishment, achieved through teamwork, underscores my ability to bring about meaningful change through strategic thinking, technological innovation, and effective leadership. Moreover, it stands as a testament to the power of teamwork, reinforcing my belief in the importance of humility and the value of every individual's contribution in achieving common goals.

Please give an example of a time when you had to overcome adversity on your way to achieving a goal.

In reflecting on my experiences, two examples readily come to mind when it comes to overcoming adversity on my path to achieving goals. As a Thermodynamics Teacher's Assistant, I encountered the challenge of updating course materials and creating extra exercises under a tight deadline. The task seemed overwhelming at first, requiring the incorporation of complex concepts from the textbook into engaging PowerPoint slides and assignments. However, through effective planning, seeking feedback, and iterating based on student input, I successfully delivered comprehensive course materials that enhanced student understanding. This experience taught me the value of resilience, effective planning, and collaborative problem-solving.

As the Lab Manager at ARTS-Lab, I faced significant resistance when introducing a machine learning model to track working hours and restructuring the lab for improved efficiency. To overcome this adversity, I took a proactive approach. I held a team meeting, actively listened to concerns, addressed privacy worries, and incorporated team feedback into the lab restructuring. Through open communication, empathy, and teamwork, we successfully implemented these changes, resulting in a more efficient lab and a stronger, more cohesive team. This experience taught me the importance of effective communication, empathy, and the power of teamwork in overcoming resistance to change.

These examples demonstrate my ability to navigate challenges, adapt to new situations, and foster collaboration in the face of adversity. They exemplify my determination, problem-solving skills, and resilience, all of which I carry forward as valuable lessons in my personal and professional journey.

What are your plans for Summer 2024? Have you accepted an internship or full-time offer of employment?

For Summer 2024, I am actively pursuing new opportunities and exploring potential internships or full-time positions that align with my career aspirations. While I have not yet accepted an offer of employment, I am proactively seeking a challenging and rewarding position where I can contribute my skills and continue to grow both personally and professionally.

During this period, I aim to leverage my diverse experiences and skill set to make a meaningful impact in a dynamic work environment. I am particularly interested in opportunities that allow me to apply my expertise in areas such as mechatronics, thermodynamics, and real-time systems or anything related to one of my majors. My goal is to work alongside experienced professionals and collaborate on projects that push the boundaries of innovation and make a positive difference in society.

In my search for the ideal position, I am actively networking, attending career fairs, and engaging with industry professionals to explore potential opportunities. I am also continuously expanding my knowledge and honing my skills through relevant coursework, research projects, and extracurricular activities.

While I am excited about the possibilities that Summer 2024 holds, I am committed to carefully evaluating each opportunity to ensure the best fit for both my career goals and the organization's objectives. I am eager to contribute my passion, dedication, and expertise to a team that values innovation, collaboration, and continuous growth.

What do you know about Chevron Corporation and the Energy industry? (Highlight your interest in working for our corporation/industry, and any particular positions of interest)

I have conducted *SOME* research on Chevron Corporation and the energy industry, and I am genuinely impressed by the company's reputation and its contributions to the global energy landscape already. Chevron's commitment to excellence, safety, and environmental responsibility resonates with me, and I find myself drawn to the exciting opportunities and challenges within the energy sector.

The energy industry has captivated my interest due to its crucial role in powering economies and driving technological advancements. It is a constantly evolving field that demands continuous innovation and sustainable practices. Chevron's global presence and diverse range of projects across the energy value chain offer a unique platform to contribute to meaningful solutions on a global scale.

As I explore potential opportunities within Chevron, I am particularly interested in roles that allow me to leverage my technical background in electronics, mechatronics, and thermodynamics, along with my experience in real-time systems and engineering project management. Chevron's emphasis on innovation, technology, and sustainability aligns with my own values and aspirations. I believe that my skills and expertise in these areas can be effectively utilized to address the energy challenges of the future.

Furthermore, I am impressed by Chevron's focus on employee development, diversity, and fostering a collaborative work culture. I recognize the importance of learning from and collaborating with talented professionals in order to continuously grow and make a meaningful impact. Chevron's inclusive environment and commitment to teamwork resonate strongly with my own values, as I implement many of the same values into the ARTS-Lab.
